

ENTREPRENEURSHIP: (SO) WHAT?

LECTURE 1

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WHAT IS ENTREPRENEURSHIP?

HOW WE DEFINE ENTREPRENEURSHIP

Entrepreneurship is the pursuit of opportunity beyond resources controlled.

– HOWARD STEVENSON

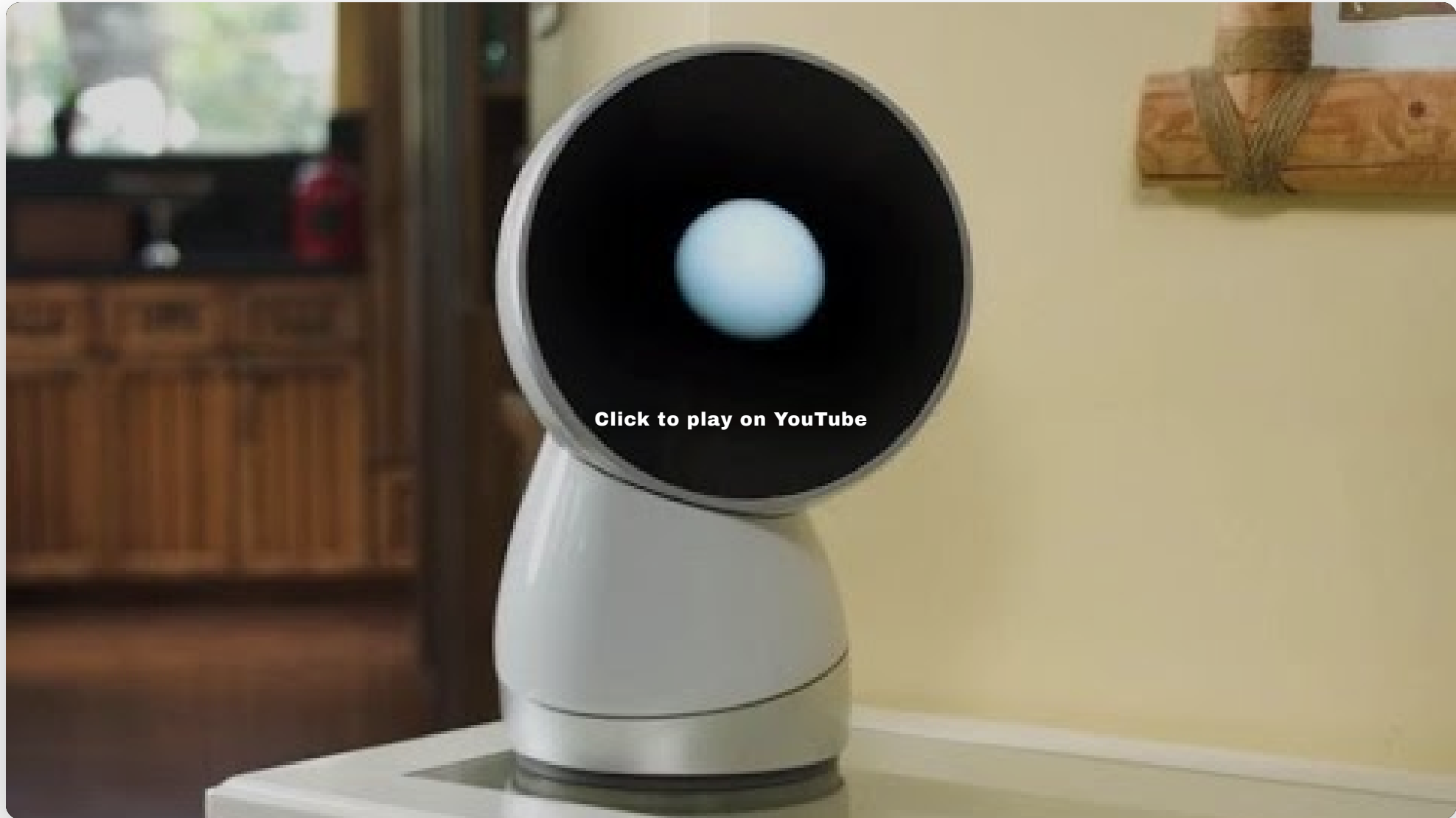
- Focus on **decision making under uncertainty** with limited resources.

$$E(V) = P \times V$$

... but P (probability of success) and V (value of a successful project) are both **unknown** up front.

- Should I pursue an opportunity (or not)? How can I learn about its value?
- Which choices do I need to make to realise the potential of this opportunity?

CASE: JIBO



CASE: JIBO

Why does Jibo fit our definition of entrepreneurship?

- The potential value of an idea (opportunity) — and how to realise it — is **uncertain up front**.
- There is **more than one path** to create and capture value from this idea.
- Only through **experimentation** can entrepreneurs learn the true value of an idea and how to realise its potential.
- Experimentation is **costly** and entrepreneurs have only limited resources — many need **external resources**.

THREE FUNDAMENTAL PROBLEMS OF ENTREPRENEURS

- 1 Entrepreneurs face **greater uncertainty**.
- 2 Entrepreneurs face **highly-skewed outcomes**.
- 3 Entrepreneurs are **highly resource-constrained**.

ENTREPRENEURS FACE GREATER UNCERTAINTY

DEMAND & MARKET

Will customers want it — and at what price?

TECHNOLOGY

Can it actually be built — and will it work?

EXECUTION

Can the team deliver, hire, and scale?

FINANCING & COST

Can we afford to find out?

ENTREPRENEURS FACE GREATER UNCERTAINTY



VANMOOF S2

New product, new(ish) firm, new category



GAZELLE ULTIMATE C380

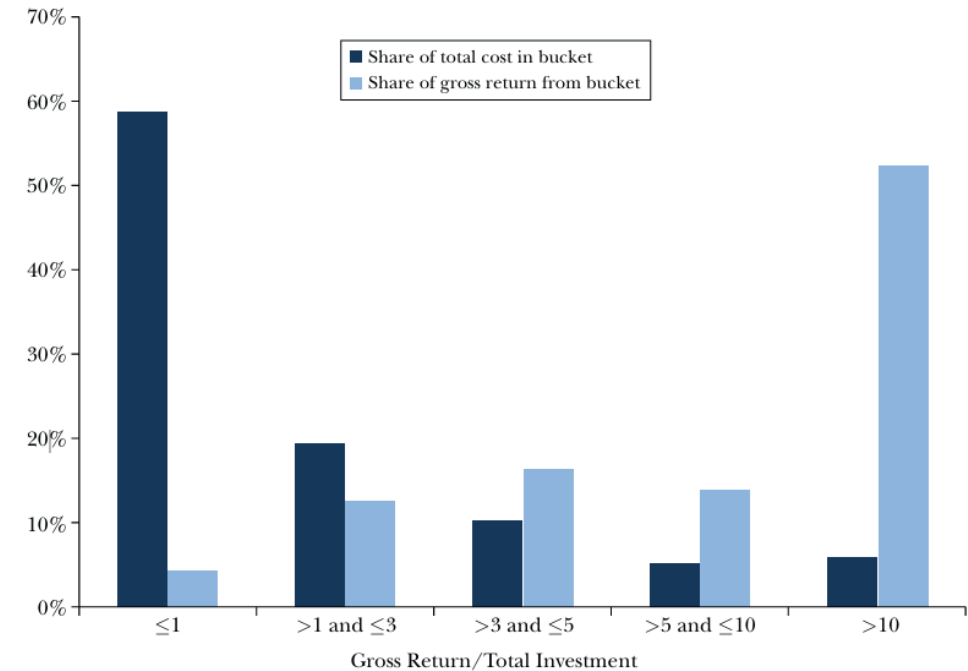
Established brand, established product

ENTREPRENEURS FACE HIGHLY-SKEWED OUTCOMES

At this major VC, almost **60%** of investments fail; only **~6%** are big successes.

Figure 1

Total Cost and Total Return for a Venture Capital Firm



Source: Authors using proprietary data from an anonymous venture capital firm that has invested more than \$1 billion over the last decade.

Notes: Investments are placed into buckets according to their return multiple. For each bucket, we show the total cost and total return for that group of investments.

ENTREPRENEURS ARE HIGHLY RESOURCE-CONSTRAINED

HUMAN RESOURCES

Only a small founding team — often a solo entrepreneur — at the start.

FINANCIAL RESOURCES

Entrepreneurs are typically cash-constrained.

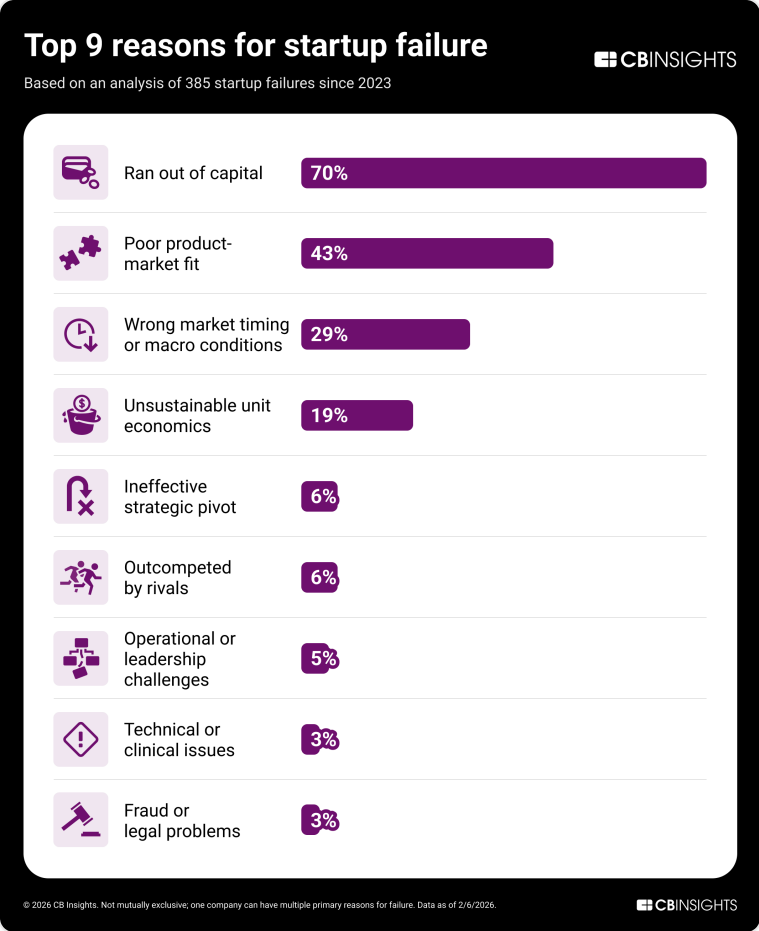
PHYSICAL CAPITAL

Lack of high-end equipment, production facilities, ...

INTANGIBLE CAPITAL

IP, goodwill, brand value yet to be built.

ENTREPRENEURS ARE HIGHLY RESOURCE-CONSTRAINED



Source: CB Insights – Top reasons startups fail

THE ENTREPRENEURIAL CATCH-22

- Uncertainty and risk can only be reduced by **having resources**.
- Resource providers might only commit when **uncertainty and risk are low**.
- Traditional business strategies and tactics **break down**.



THE QUESTIONS THIS RAISES

PART 1 OF THE COURSE

- Who becomes an entrepreneur and why, when the odds of success are unknown but likely very low?
- What are the private and social returns to entrepreneurship?
- Which individual and contextual factors impact entrepreneurial performance?

PART 2 OF THE COURSE

- How do entrepreneurs form their initial beliefs about opportunities?
- How do they learn in an uncertain environment with limited resources to experiment?
- How do they ultimately persuade others?

If we want to know the answer to many of
these questions, we first need to know...

HOW TO MEASURE ENTREPRENEURSHIP?

WHICH COUNTRY IS MORE ENTREPRENEURIAL?

A

US
UNITED STATES

B

EC
ECUADOR

C

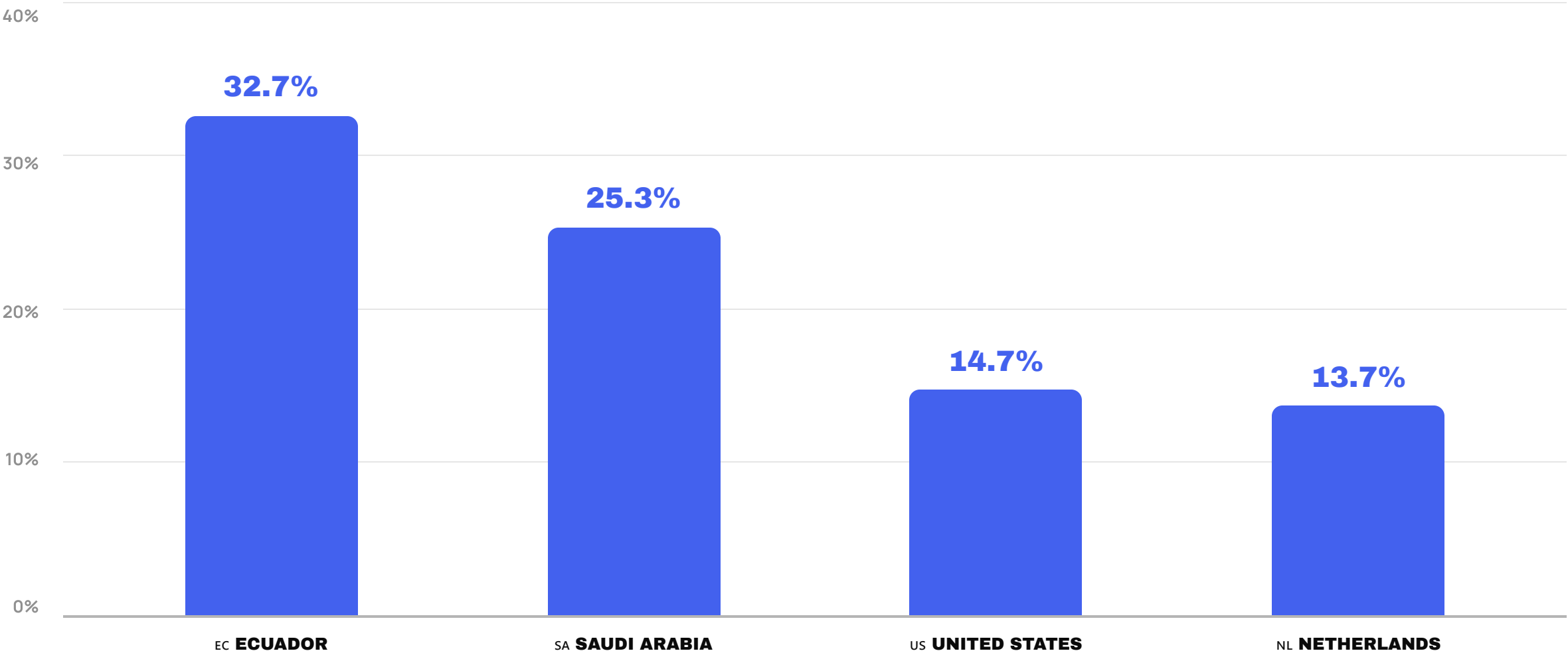
SA
SAUDI ARABIA

D

NL
NETHERLANDS

TOTAL EARLY-STAGE ENTREPRENEURIAL ACTIVITY

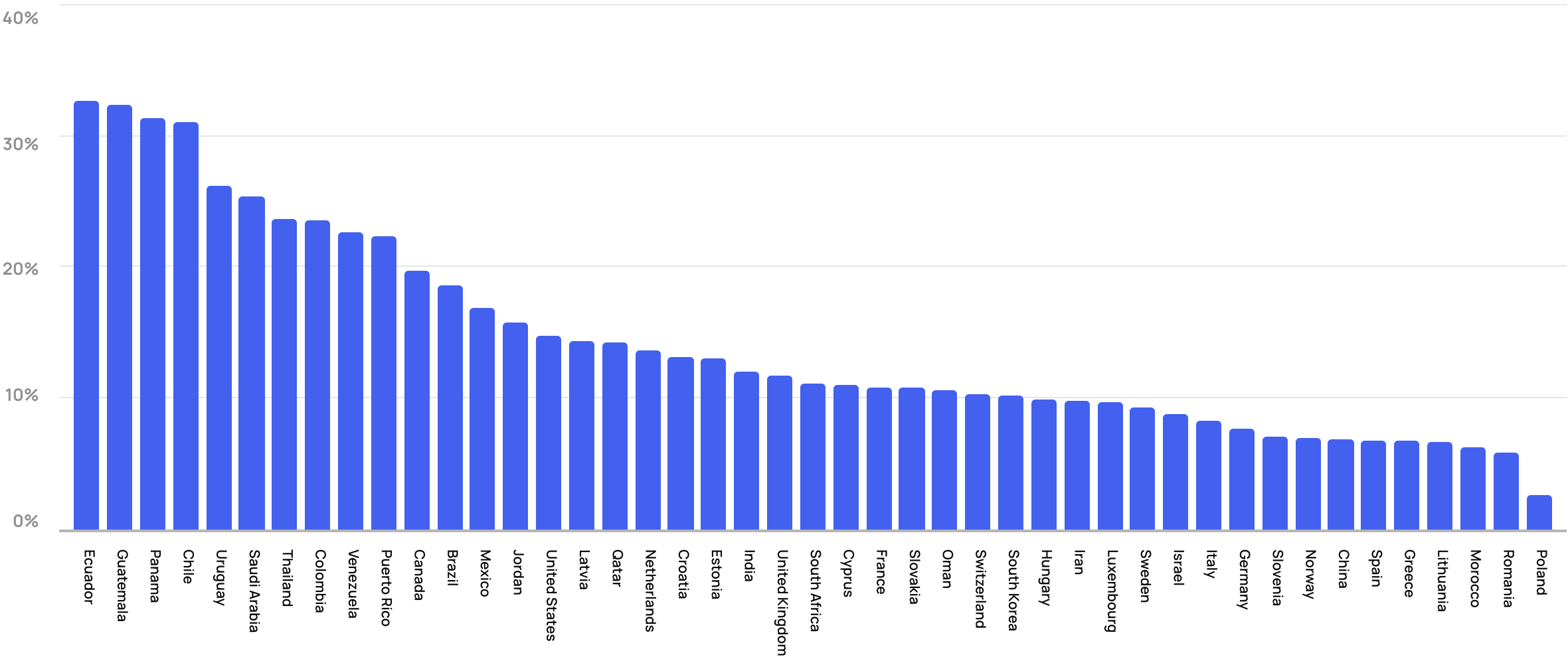
Share of adults aged 18–64 who are starting or running a new business (TEA)



Source: Global Entrepreneurship Monitor (GEM), 2023

TOTAL EARLY-STAGE ENTREPRENEURIAL ACTIVITY

All 45 economies in the 2023 GEM survey, ranked



Source: Global Entrepreneurship Monitor (GEM), 2023

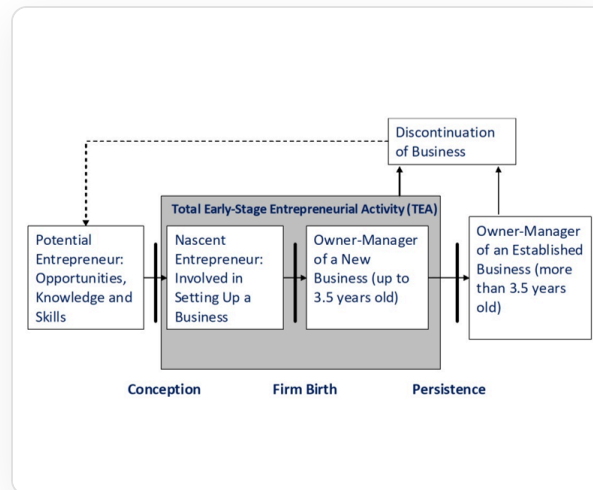
WHAT IS AN ENTREPRENEUR (ACCORDING TO THE GEM)?

Entrepreneur

An adult who is engaged in setting up or operating a new venture that is ≤ 42 months old.

TEA rate

The percentage of the 18–64 population who are either a **nascent entrepreneur** or an **owner-manager of a new business**.

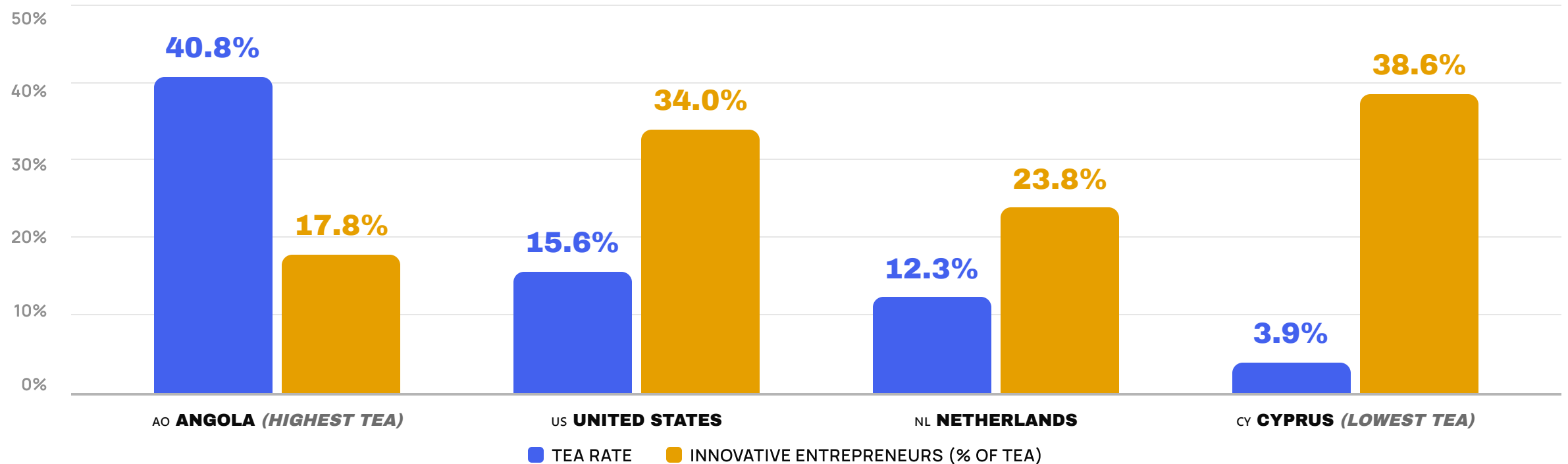


PROBLEMS WITH USING NEW VENTURE CREATION AS DEFINITION OF ENTREPRENEURSHIP

- 1 Many new ventures are mundane, hobby businesses that generate little private or social value — “**subsistence**” rather than “**transformational**” entrepreneurship.
- 2 A focus on new firms (≤ 3.5 years) excludes the **growth and exit stages** of entrepreneurship.
- 3 Excludes people who are in the entrepreneurial process but never reach the stage of actually setting up a business (**survival bias**) — roughly **50%** of everyone who considers starting a business (Bennett & Chatterji, 2023).

SUBSISTENCE OR TRANSFORMATIONAL ENTREPRENEURSHIP?

2018 TEA rate vs. share of TEA who are innovative entrepreneurs

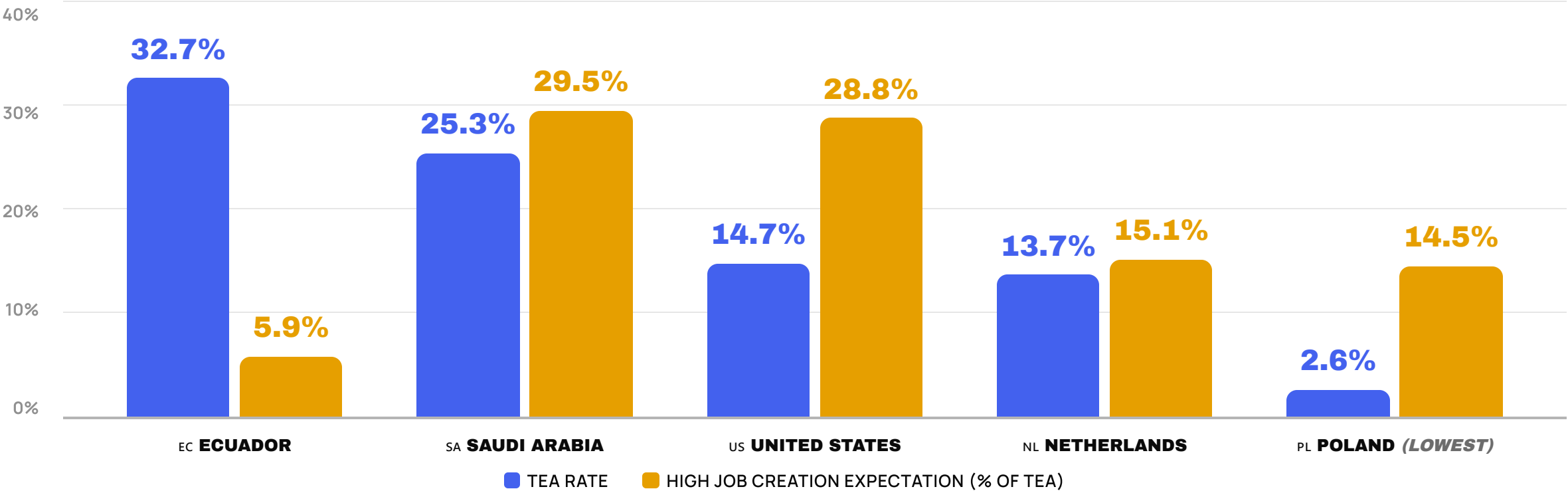


Innovative entrepreneur (GEM) – share of those involved in TEA who indicate that their product or service is new to at least some customers *and* that few or no other businesses offer the same product.

Source: Global Entrepreneurship Monitor (GEM), 2018 – Angola substitutes for Ecuador, which was not surveyed in 2018

SUBSISTENCE OR TRANSFORMATIONAL ENTREPRENEURSHIP?

2023 TEA rate vs. share of TEA with high job creation expectation

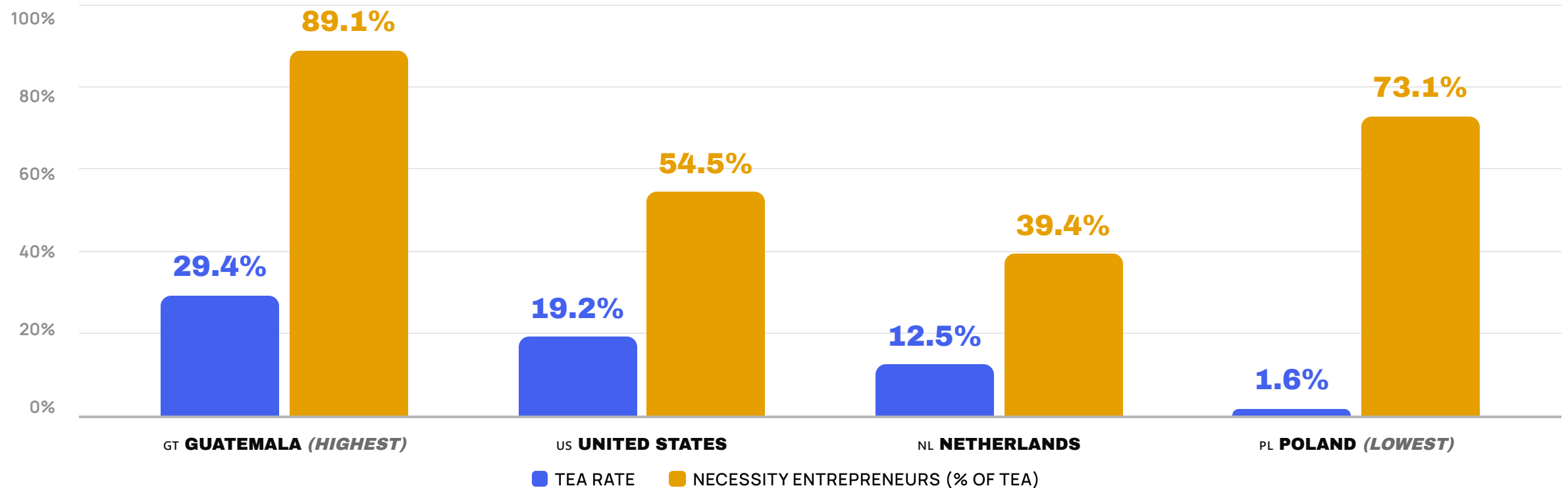


High job creation expectation (GEM) — share of those involved in TEA who expect to create 6 or more jobs in the next 5 years.

Source: Global Entrepreneurship Monitor (GEM), 2023

SUBSISTENCE OR TRANSFORMATIONAL ENTREPRENEURSHIP?

2022 TEA rate vs. share of TEA who are **necessity** entrepreneurs



Necessity entrepreneur (GEM) — share of TEA who responded that they became an entrepreneur “to earn a living because jobs are scarce” (variable TEA22M0T4yes).

Source: Global Entrepreneurship Monitor (GEM), 2022

NOT EVERYONE WHO CONSIDERS ENTREPRENEURSHIP SETS UP A NEW BUSINESS

Global GEM 2022 – weighted across all 49 surveyed economies

GENDER	TEA RATE	ENTREPRENEURIAL INTENTIONS
♂ MALE	11.8%	20.1%
♀ FEMALE	7.7%	14.7%
OVERALL	9.7%	17.3%

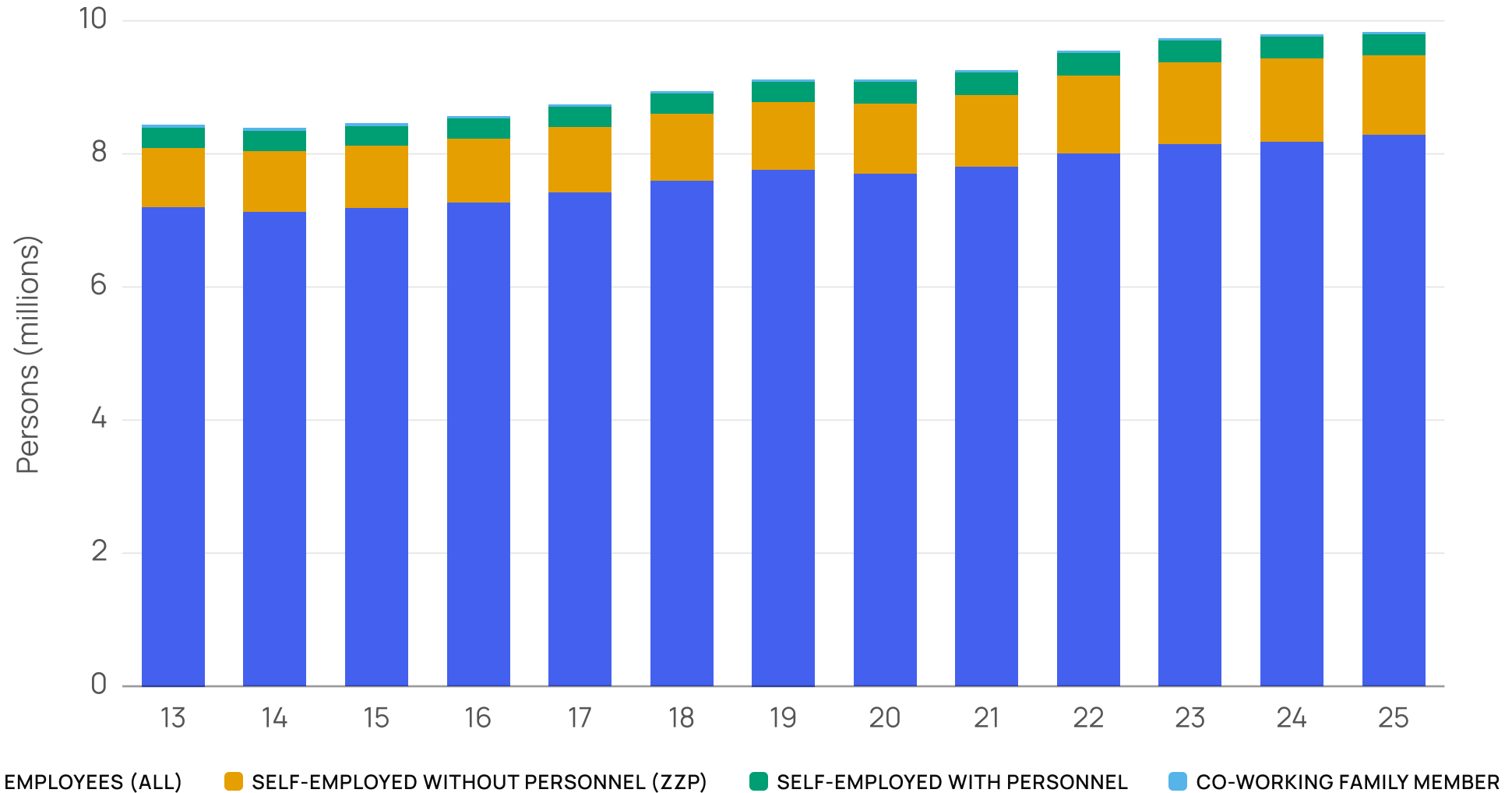
Entrepreneurial intentions (futsup) – share of the 18–64 population (excluding those already engaged in entrepreneurial activity) who intend to start a business within the next three years.

Source: Global Entrepreneurship Monitor (GEM) Adult Population Survey, 2022 – computed from individual-level data (weighted), n ≈ 172,000 respondents.

IS ENTREPRENEURSHIP = SELF-EMPLOYMENT?

- Self-employment captures the **income risk-taking** aspect of entrepreneurship.
- Easy to track — a **measure available at the individual level**, in administrative data and labour-force surveys.

IS ENTREPRENEURSHIP = SELF-EMPLOYMENT?



Source: CBS, Werkzame beroepsbevolking; positie in de werkring (×1,000 persons)

IS ENTREPRENEURSHIP = SELF-EMPLOYMENT?

- **False self-employment** — someone carries out an assignment as a self-employed professional but is in reality an employee — e.g. they cannot decide themselves when or how to do the work. The client is effectively the employer but pays no payroll taxes.
- **Management companies (Belgium)** — holdings set up mainly by high-income individuals so income is taxed as corporate earnings instead of as wages (and many purchases can be deducted as operating costs).



THE SELF-EMPLOYED ARE A MIXED BUNCH

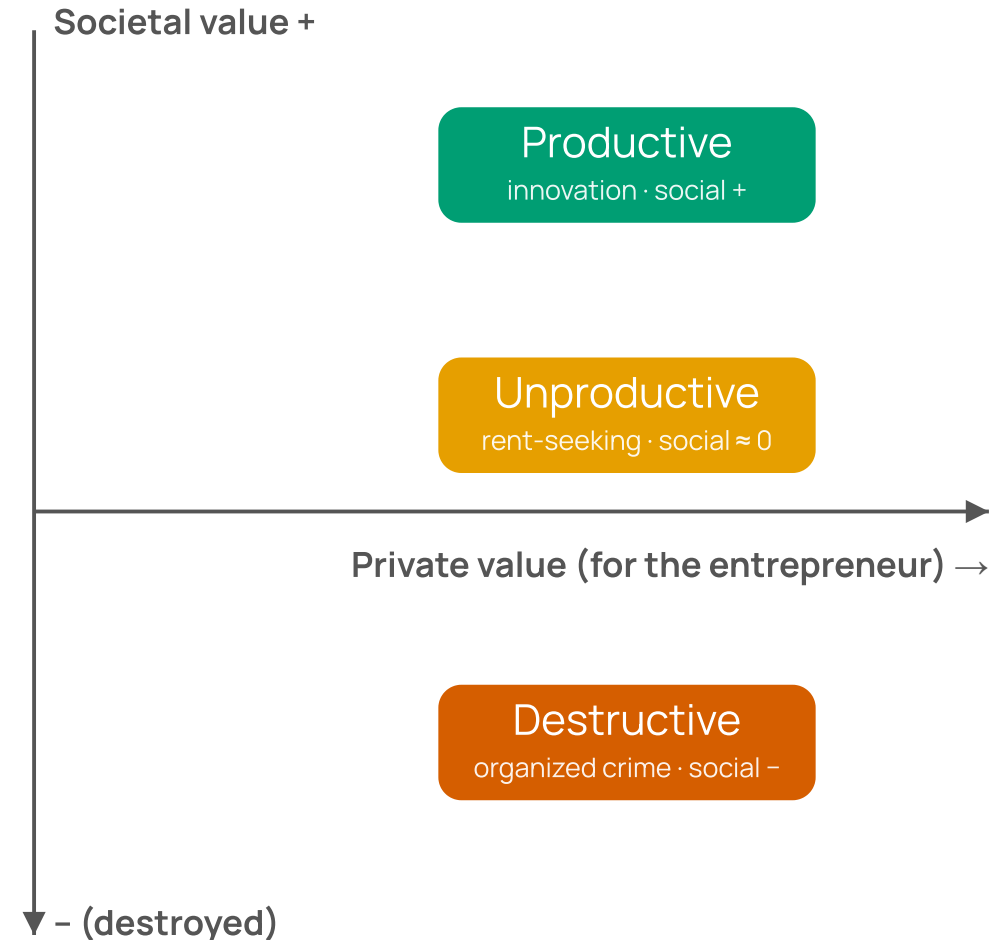
	All	Salaried	Self-employed		
			All	Unincorp.	Incorp.
Observations	1,225,886	1,108,591	117,295	75,476	41,819
	100.0%	90.4%	9.6%	6.2%	3.4%
A. LABOUR MARKET OUTCOMES					
Mean earnings	\$47,515	\$46,421	\$58,174	\$40,820	\$89,169
Median earnings	\$36,090	\$36,363	\$34,190	\$24,625	\$55,591
Median hourly earnings	\$18.0	\$18.0	\$17.4	\$13.8	\$24.6
Annual hours worked	1,985	1,976	2,078	1,936	2,331
Full-time, full-year	0.69	0.70	0.64	0.57	0.78
B. DEMOGRAPHICS					
Age	40.2	40.0	42.9	42.4	43.6
White	0.70	0.69	0.79	0.76	0.83
Female	0.48	0.49	0.36	0.40	0.28
Years of schooling	13.7	13.7	13.9	13.6	14.5
College graduate (or more)	0.33	0.33	0.36	0.31	0.46

Source: [Levine & Rubinstein \(2017\)](#), “Smart and Illicit: Who Becomes an Entrepreneur and Do They Earn More?”, QJE 132(2), Table I, Panel A (CPS 1996–2012).

PRODUCTIVE, UNPRODUCTIVE, AND DESTRUCTIVE ENTREPRENEURSHIP

“The basic hypothesis is that, while the total supply of entrepreneurs varies among societies, the productive contribution of the society’s entrepreneurial activities varies much more because of their allocation between productive activities such as innovation and largely unproductive activities such as rent seeking or organized crime. This allocation is heavily influenced by the relative payoffs society offers to such activities.”

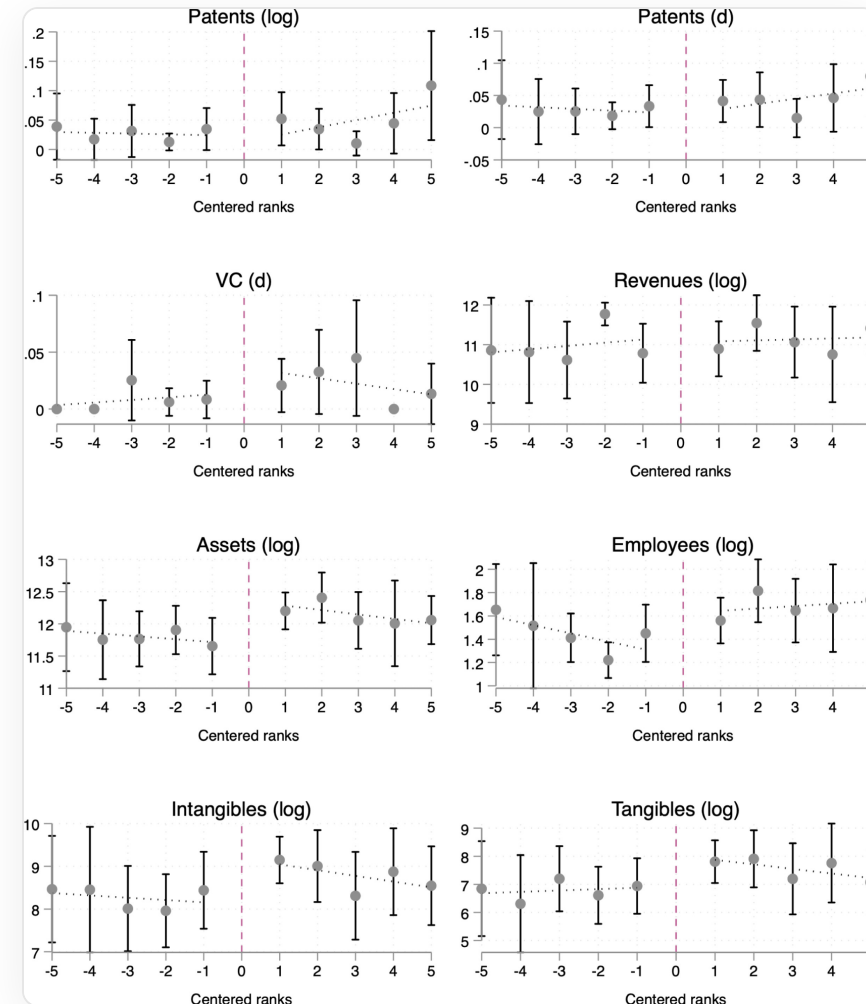
— Baumol (1990), *Journal of Political Economy* 98(5), 893–921



DO SUBSIDIES SPUR PRODUCTIVE OR UNPRODUCTIVE ENTREPRENEURSHIP?

*“We document a **null effect** of [local public programs targeting innovative start-ups] over a wide range of firm-level outcomes. However, we find that securing local subsidies increases start-ups’ probability to obtain **additional** public subsidies, which points in the direction of a vicious ‘Matthew effect’ in subsidy allocation.”*

— Santoleri & Russo (2025), *Research Policy* 54(1)

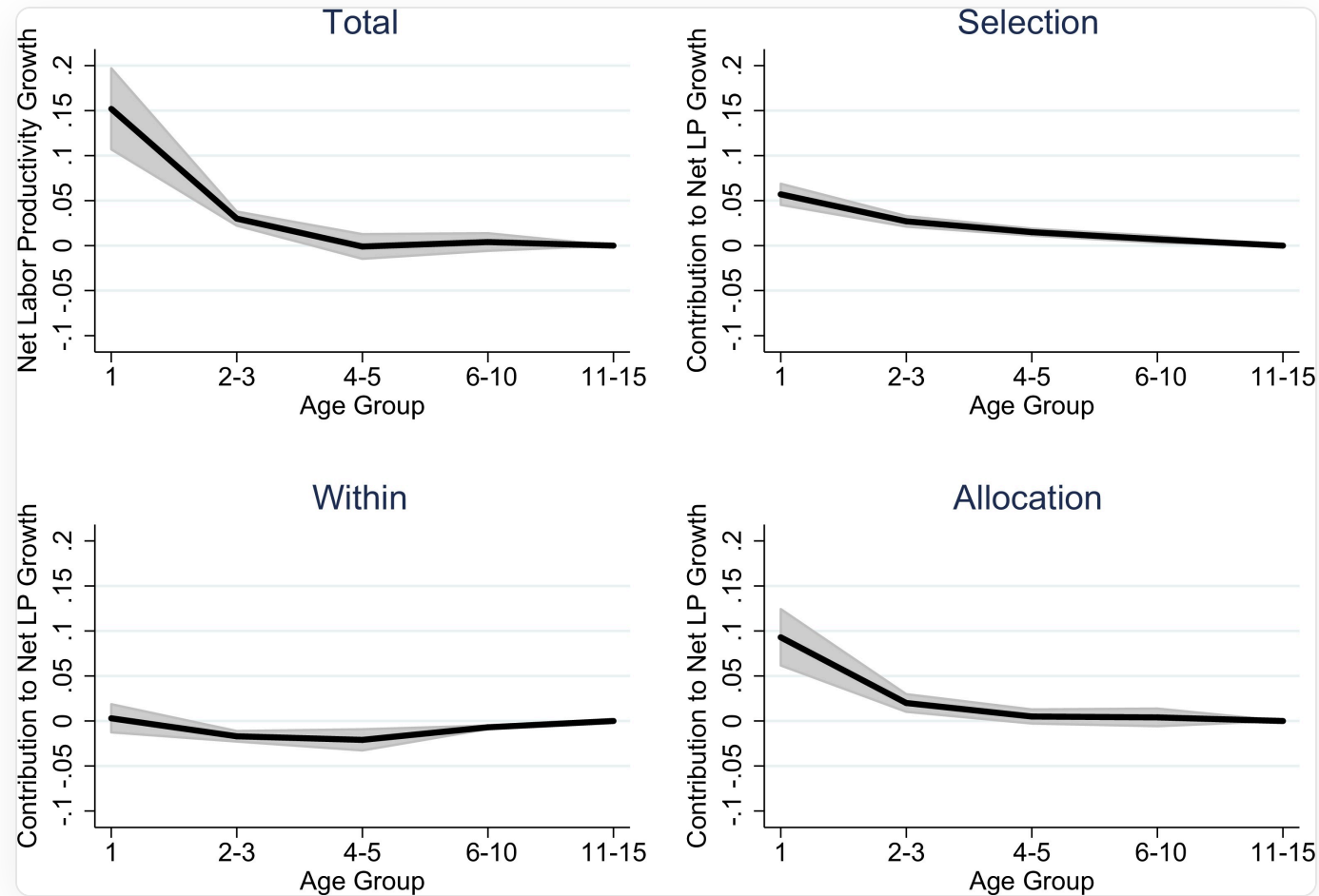


IN A NUTSHELL: MEASURING ENTREPRENEURSHIP IS HARD

- Entrepreneurship is a **wide spectrum** — from bootstrapped software apps to VC-backed deep-tech start-ups in the space industry.
- Operationalising all the distinctive characteristics (*uncertainty, limited resources, experimentation*) at once is very challenging at scale.
- Each definition is a **trade-off**:
 - Restricting to start-ups with an innovative product leaves out entrepreneurs who *tried* to create something innovative but failed along the way.
- Keep these trade-offs in mind throughout the course — in your own research, when reading policy reports, in the news, ...

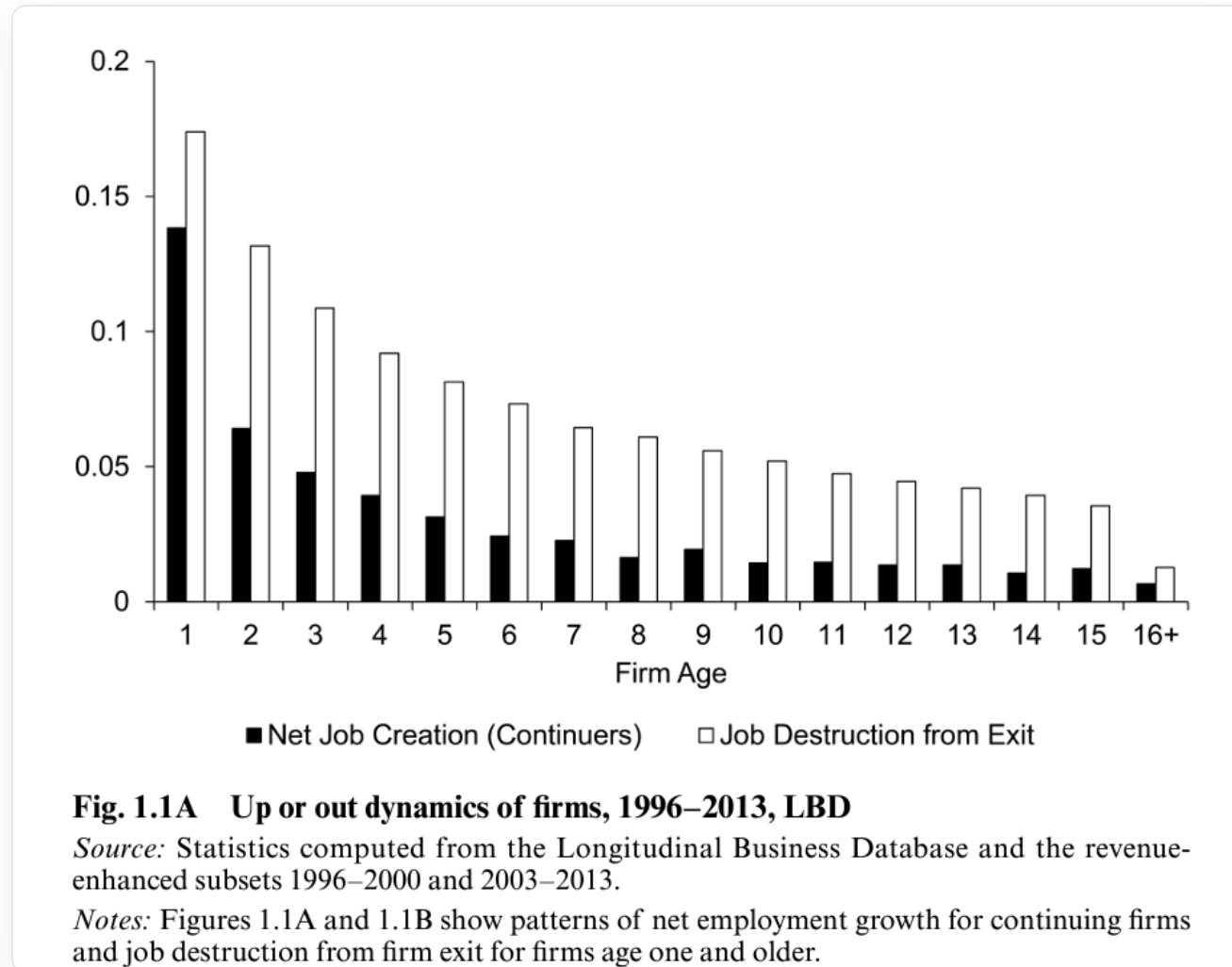
**WHY SHOULD WE
CARE ABOUT
ENTREPRENEURSHIP?**

STARTUPS CONTRIBUTE MOST TO PRODUCTIVITY GROWTH

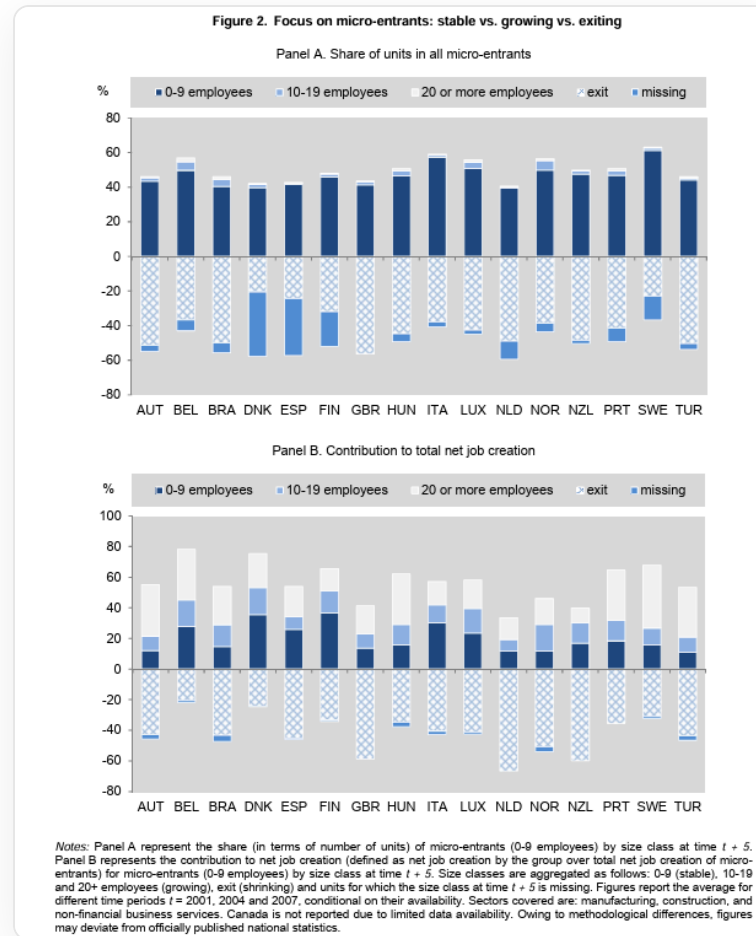


Source: Alon, Berger, Dent & Pugsley (2018), "Older and slower: The startup deficit's lasting effects on aggregate productivity growth", *Journal of Monetary Economics* 93, 68–85.

STARTUPS CREATE (AND DESTROY) MOST JOBS



... BUT ONLY A SMALL FRACTION OF STARTUPS CREATE ALL THOSE JOBS

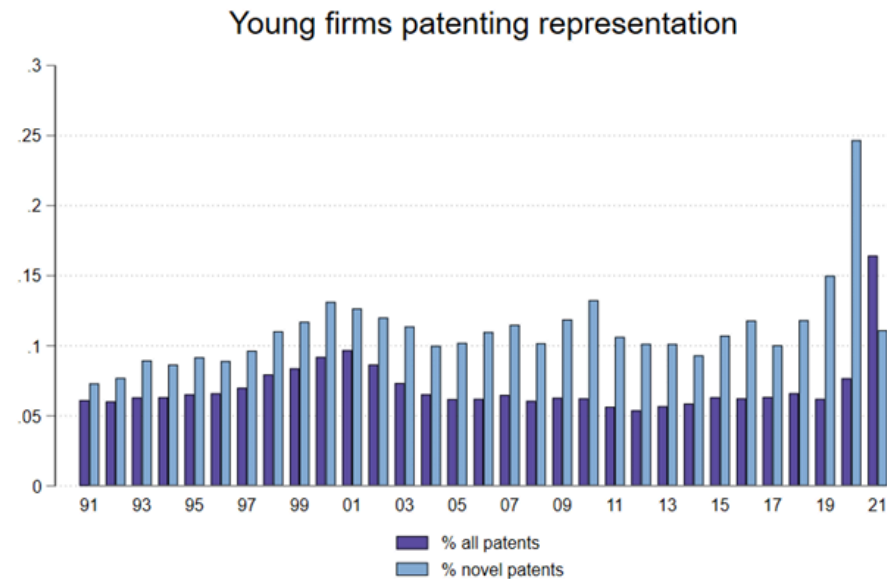


Source: Calvino, Criscuolo & Menon (2016), "No Country for Young Firms?", OECD STI Policy Papers No. 29, OECD Publishing.

STARTUPS DISPROPORTIONATELY CONTRIBUTE TO BREAKTHROUGH INNOVATIONS

Figure 5: Percentage of patents and novel patents by young firms over time

Notes: The figure reports the fraction of all patents that are patented by firms less than 3 years old (% all patents). It also reports the fraction of all novel patents (i.e., those patents that have the first reference to a IPC section, class, and subclass) whose assignee is less than three years old at the time of the patent application.



SLIDES & DATA

The slides and the data behind every graph in this lecture are available at:

github.com/jmahieu/foundations-of-entrepreneurship